Southern New Hampshire University's Computer Science program has five elements that they have deemed the most important skills for someone to graduate with. They are

- Collaborating in a Team Environment
- Communicating with Stakeholders
- Data Structures and Algorithms
- Software Engineering and Database
- Security

The first few courses focused on the Software Engineering side of things. We learned how to make a project in Python that collects information about different monkey rescues along with being able to display that information. It did not require a database as the data was a lower amount. This taught me how to start a project and finish it within the requirements set by the "stakeholders."

Each course had an element of collaborating in a team environment as we did discussion posts and code reviews on each other's work. The course taught you to set aside your ego on being critiqued and the value of having people work with you. I found this similar to my previous job being a teacher. A lot of teaching is about working together with your administrator and your coworkers. There are a lot of requirements that are explained rapidly, and you must adapt to them quickly. This need for adaptation and cooperation goes for Computer Science as well.

Data Structures and Algorithms were explored in many different programming languages. Arrays, LinkedLists, Hash Tables, Binary Search Trees were all covered in a course where you had to code the various structures yourself. This course also focuses on the need to learn which structure has what advantages. Certain structures are better for different tasks.

Database courses were the ones I enjoyed the most. We looked into SQL and non-SQL databases with Mongo being the most common noSQL database. This included a project which combined the use of Python and MongoDB to create a dashboard. This is one of the artifacts in the portfolio. Another course was focused on the use of a csv file being converted into SQL tables that could then be joined for research use. Finally, there was a gap analysis using PowerBI to display the inventory and use of various goods from a shop. The goal was to see what the data showed before using PowerBI to display and explain the data along with how the data could be improved.

Security was emphasized in every course. Every project we were reminded to do input validation, authentication, authorization, and obfuscation of the inner workings. One of the biggest skills of Object-Oriented Programming is Encapsulation which is the ability to obscure inner workings of a project from the end user. An example of this is using a phone you have access to the apps and screen, but not the springs and chips inside the phone.

Below are explanations of each artifact present in my protfolio.

Data Structures and Algorithms

Language(s): C++

Project: Sorting Courses from a csv file to a Binary Search Tree

Folder Contents: README.md, VectorSorting.cpp, LinkedList.cpp, HashTable.cpp, ProjectTwo.cpp, Data Structures and Algorithms Graphic, and Enhancement Narrative.

This course focused on learning how to create various data structures, and on what algorithms are best used in different situations. We learned about Big O notation and the efficacy of the different algorithms. We also learned how to code the data structures in terms of working with algorithms for practical applications.

Databases

Language(s): SQL

Project: Gap Analysis on shipping and returns for a business

Folder Contents: README.md, FleetMaintenanceRecords.csv, rma.csv, Gap Analysis, Enhancement Narrative

This course focused on the use of mySQL to read in a csv file to tables and then use the created tables to analyze the reports on return shipments. To enhance this, I have done a gap analysis to go further into the data and discuss the possibilities with the data currently in use and what more could be done with expanded data.

Software Design

Language(s): Python, pyMongo, Jupyter Notebook

Project: Animal Search and Rescue Shelter Dashboard

Folder Contents: README.md, ProjectTwoDashboard, animalShelter, Enhancement Narrative, and logo

This course focused first on the use of Python and pyMongo to created CRUD functions in python to be used for a dashboard. The course then moved towards creating a dashboard in which a user cna look at different serach and rescue animals from animal shelters to be used and rented out. The dashboard includes sorting options for different rescue types, sorting for dog vs cat, and the ability to see on a map where the animal is located.